RENALIS

A dialysis twin that makes competing explanations and the next evidence question visible

Research prototype

Historical records + available device observations + explicit uncertainty

The clinical research question

A declining pressure trace can fit several physiological explanations

During dialysis, fluid removal interacts with refill and compensatory responses.

RENALIS preserves several compatible models and compares hypothetical profiles.

The next question is which feasible observation could change that comparison.

One patient, one evidence cutoff

Six connected stages keep the reasoning visible

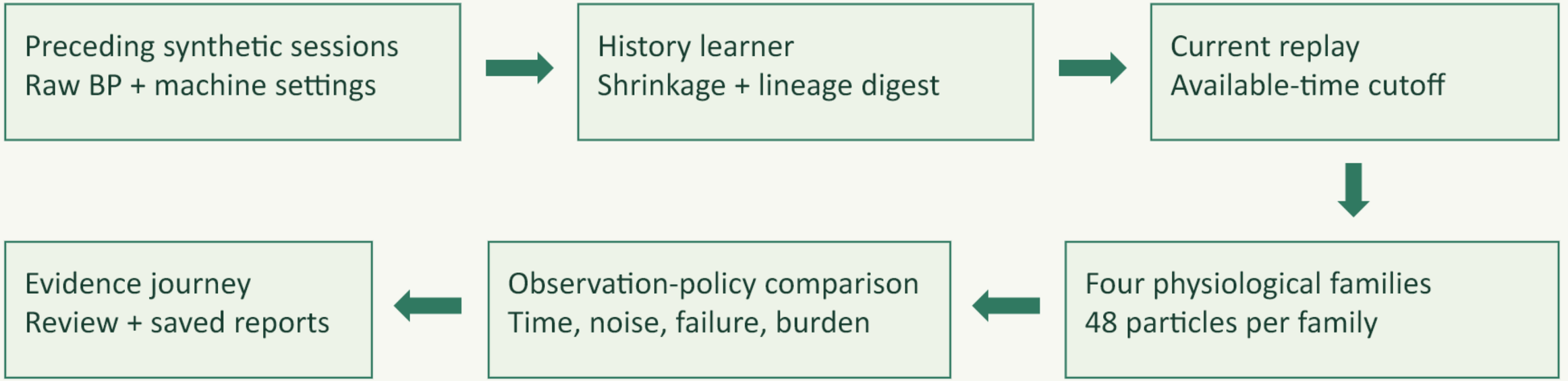
Learn from preceding sessions, then inspect the model state.

Compare four explanations, then preview possible measurement results.

Acquire or abstain, review the record, and inspect independent evaluations.

Software architecture

The synthetic physiological path and observed-data benchmark remain separate



Patient history changes the prior

Raw preceding-session trajectories replace an authored refill prior

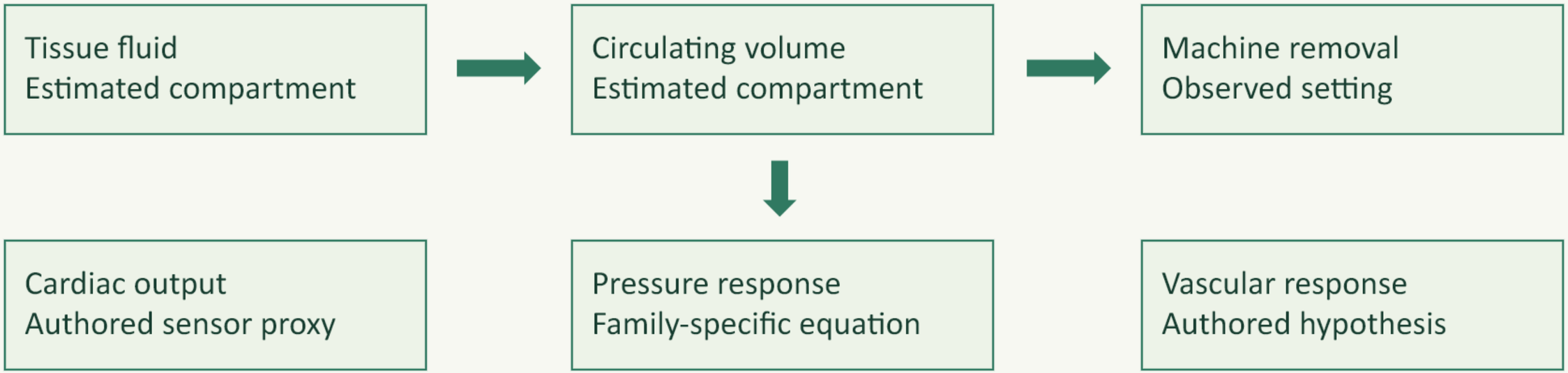
Six completed synthetic sessions supply BP readings and machine settings.

A grid likelihood fits refill and compensation, then shrinks toward a population reference.

A forward-session ablation, uncertainty floor and history digest expose the learning limits.

Inspectable physiological structure

Compartment state, measured settings and authored response maps are labeled separately



Four explanations preserve ambiguity

Volume/refill, vascular, cardiac and mixed response families

Each family retains 48 particles and fits the same available BP and machine history.

Compatibility weights describe this chosen model bank and can remain unresolved.

Contradictory observations downgrade authority because actual physiology may be absent.

Possible measurements appear before acquisition

Predicted ranges include selected noise and observation timing

Cuff BP, relative blood volume and cardiac output have explicit observation equations.

Lower, middle and higher result previews show conditional support and scenario changes.

Overlapping ranges reveal when a measurement cannot reliably separate explanations.

Whole-policy observation value

The comparison includes what happens while waiting and when acquisition fails

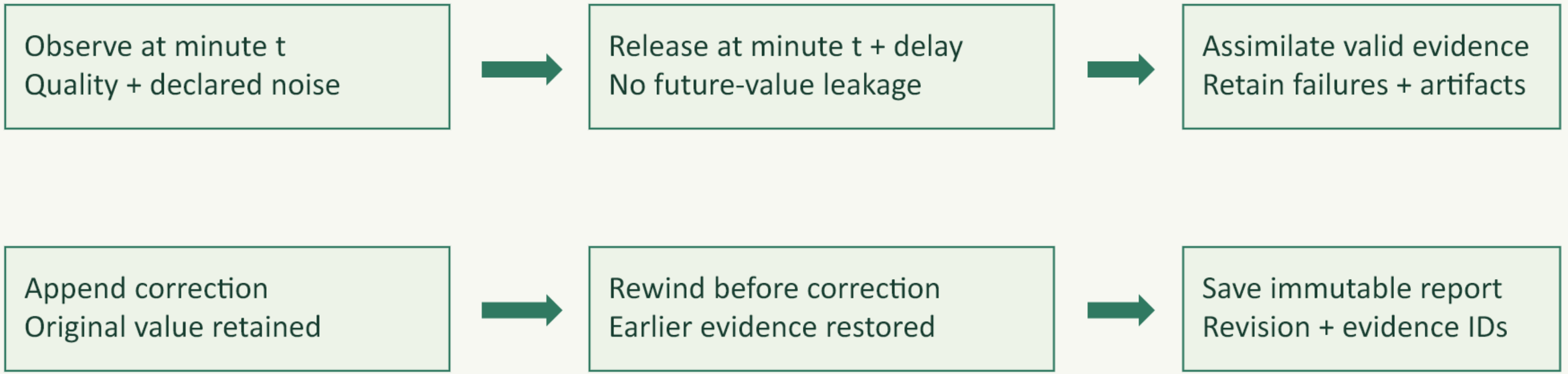
Start with the best immediate action under the declared research loss.

Integrate possible results, delayed action and acquisition failure using Gaussian quadrature.

Subtract declared burden and delay penalty. Preserve negative value and abstain when needed.

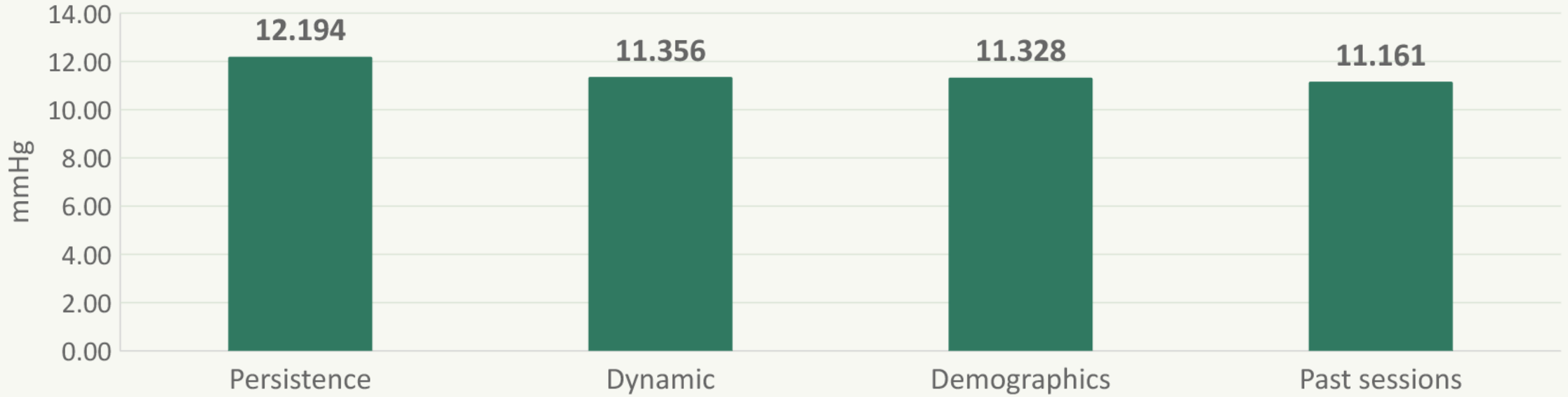
An auditable evidence lifecycle

Availability, quality, corrections and saved revisions share one contract



History adds a small retrospective gain

HEMOBP v3: 61,947 test windows from 176 separate test patients



History increment: -0.1665 mmHg. Patient-cluster interval [-0.2298, -0.1099].

The observed-data limits change interpretation

The clock-alignment assumption and alert workload remain visible

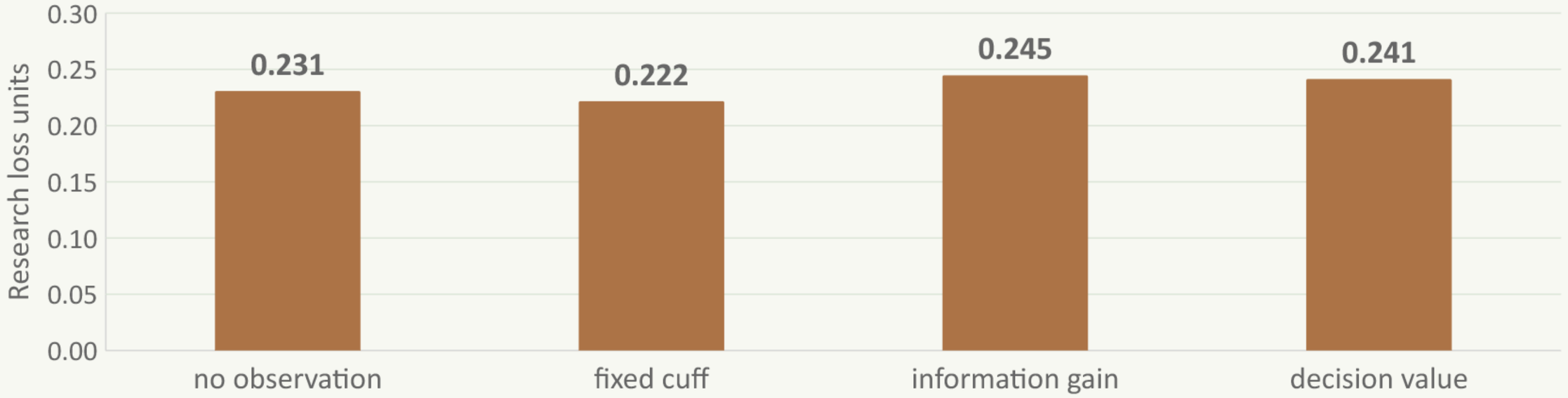
An empirical eight-hour table offset matches 2,516,989 of 4,366,298 machine packets.

Source clock semantics still need independent confirmation. Exclusions may bias the cohort.

Sensitivity is 61.25%, PPV is 13.51%, and empirical 90% range coverage is 86.74%.

Decision value does not beat cuff overall

Independent authored holdout: 192 cases across eight regimes



Decision value minus cuff: +0.01975, paired seed interval [-0.00981, 0.05996].

Failure states remain part of the product

A clear demonstration includes the cases where the model should stop

Current low BP, stale readings, completion and contradiction withhold scenario authority.

Failures and artifacts remain visible. Equipment constraints apply on the server.

Network errors disable mutations and offer retry. Reports preserve evidence through corrections.

Longitudinal and interoperability experiments

Both extensions expose their assumptions

Conditional fluid transport spans future sessions with authored intake/output and conservation checks.

A synthetic FHIR R4 collection exports Patient, Device and measured BP Observation resources.

Prospective physiology, ABDM profile conformance and authorized clinical integration remain open gates.

The next evidence gates

A research prototype becomes credible through independent evidence

Confirm source clock semantics and reproduce the audited cohort independently.

Conduct an approved expert walkthrough and paired physiological silent study.

Evaluate external calibration and workload before any prospective impact or treatment claim.

Sources and interpretation

Implementation: backend/history.py, model.py, mechanisms.py, insights.py and app.py.

Observed results: docs/observed-v04.json and scripts/evaluate_observed_v04.py.

Policy results: docs/policy-validation-v04.json. Initial report and metric amendment retained.

Dataset: Wu, C.-J., HEMOBP v3 (2018), CC BY 4.0. DOI 10.6084/m9.figshare.6260654.v3.

Data publication: https://www.nature.com/articles/s41597-019-0319-8

Clinical context: KDIGO BP and Volume Management in Dialysis conference report (2020).

FHIR R4: https://hl7.org/fhir/R4/observation.html

ABDM guide: https://nrces.in/ndhm/fhir/r4/

Research and competitor decisions: RENALIS_Research_Strategy_and_Master_Plan.pdf.

Study design: docs/clinical-study-protocol.md. No actual clinical study is claimed.

This PDF preserves reviewed slide renders. Editable charts/diagrams and per-slide notes

are in the corresponding PPTX. Software checks do not establish clinical usefulness.